



UNITED INTERNATIONAL UNIVERSITY (UIU)

Dept. of Computer Science & Engineering

Assignment 2

Course No: CSE 4325 Course Title: Microprocessor & MicroController

Marks: 20

1. Draw the timing diagram for **memory Read operation** of microprocessor 8086 showing the activities of Mem/*IO*, Address-Data bus ($AD_0 - AD_{15}$, $AD_{16} - AD_{19}$), ALE, $\overline{WR}$, $\overline{RD}$, $\overline{DEN}$, DT/R in each clock cycle. (2)
2. Consider the following fetch cycle in an 8086 BIU:

[Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch, Fetch]

When the first instruction is being executed, two instructions are fetched and saved in the instruction queue. If the 2nd instruction is a 'MOV <address>' instruction and the 5th is a 'JUMP <7th instruction address>', then draw the corresponding BIU and EU's cycle. (2)

a) Write the assembly code for the below expression using the RAM content:

$$27 + 25 \times 2 - 35 = ?$$

3. [Multiplication can be done by addition] (3)
- b) Fill-up the RAM content table with the instructions machine code. (Start from 0H). Use the opcode given in the opcode table: (3)

RAM Content :

Address	Content	Address	Content
0H		8H	
1H		9H	
2H		AH	
3H		BH	01100100
4H		CH	00011011
5H		DH	00011001
6H		EH	00010100
7H		FH	00100011

Opcode Table:

Mnemonic	Opcode
HLT	1000
OUT	0001
SUB	0011
ADD	0100
LDA	1100

- c) What are the values of Accumulator registers after executing each instruction programmed in the RAM. [Initially Accumulator = 0H]. (2)
4. Explain what changes you have to make in SAP-1 architecture to perform the instruction **LDB_{OUT}** (Load the content of memory address to B register and show its value in the binary display). Also, Write the control words for execution T-states (T4, T5, T6) to perform the instruction. (3)

Control Word of a Typical SAP-1 Architecture = $\begin{matrix} C & E & L & C \\ p & p & M & E \end{matrix}$ $\begin{matrix} L & E & L & E \\ i & i & A & A \end{matrix}$ $\begin{matrix} S & E & L & L \\ u & u & B & O \end{matrix}$

5. What are the values of Accumulator registers after executing each instruction programmed in the RAM. [Initially Accumulator = 0H]. (3)

RAM Address	RAM Content		RAM Address	RAM Content
0H	0000 1001		8H	XXXX XXXX
1H	0001 1011		9H	0010 1001
2H	0010 1010		AH	0000 0111
3H	0001 1100		BH	0001 0101
4H	1110 XXXX		CH	0000 0100
5H	1111 XXXX		DH	0011 1100
6H	XXXX XXXX		EH	XXXX XXXX
7H	XXXX XXXX		FH	XXXX XXXX

6. Draw the timing diagram for **memory Write operation** of microprocessor 8086 showing the activities of Mem/IO, Address-Data bus ($AD_0 - AD_{15}$, $AD_{16} - AD_{19}$), ALE, WR, RD, DEN, DT/R in each clock cycle. (2)